

Tyler Roberts

Senior Software Engineer

CONTACT

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SKILLS

Programming

Python, JavaScript, TypeScript, Java

Machine Learning

Machine Learning Systems, Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Model Evaluation Metrics, Prompt Engineering, Vector Databases

Backend

REST APIs, Protobuf, Celery, Distributed Systems, Event-Driven Architecture, Low-Latency Services

Cloud

AWS, Cloud-Native Architecture, Kubernetes, Docker, Serverless

Data

Relational Databases, Low-Latency Datastores, Vector Search, Data Pipelines

DevOps

CI/CD, Infrastructure as Code, Observability, Production Monitoring

EDUCATION

Florida State University

Bachelor of Computer Science
2009 - 2013

Professional Summary

Senior Software Engineer with 9+ years of experience designing, building, and operating production-grade machine learning, generative AI, and cloud-native systems. Strong background delivering ML-powered backend services, scalable APIs, and AI platforms using Python, AWS, Docker, Kubernetes, and distributed systems. Proven ability to own ML tooling, training and inference pipelines, model evaluation frameworks, and AI service integrations end-to-end in regulated and data-sensitive environments. Experienced in building retrieval-augmented generation (RAG) solutions, vector search pipelines, and AI-enabled services that support real-world decision-making and user-facing products. Adept at collaborating cross-functionally with data scientists, clinicians, product managers, and platform teams to translate complex requirements into reliable, secure, and maintainable systems. Known for a hands-on, ownership-driven approach that balances engineering quality, delivery speed, and long-term scalability.

Professional Skills

Senior Software Engineer

Web Venturez LLC | February 2020 - October 2025

- Designed and built production-grade backend services and APIs powering AI-enabled and data-driven SaaS platforms, with a strong focus on reliability, scalability, and performance.
- Implemented machine learning and generative AI integrations, including LLM-based features, RAG pipelines, and vector database-backed retrieval services used in customer-facing workflows.
- Developed training, inference, and evaluation tooling to support the full AI/ML SDLC, enabling repeatable experimentation, model validation, and controlled production releases.
- Built and maintained RESTful APIs using Python and Node.js, leveraging Protobuf for efficient service communication and clear contract definitions across distributed systems.
- Deployed and operated containerized services using Docker and Kubernetes on AWS, ensuring secure handling of sensitive data and high availability in production environments.
- Collaborated closely with product managers, data scientists, and domain experts to translate complex requirements into robust technical solutions while mentoring engineers on best practices.

Full Stack Engineer

Webiz Digital | July 2017 - January 2020

- Built and maintained full-stack web applications using Python, Node.js, and modern JavaScript frameworks, supporting data-intensive and user-facing business workflows.
- Designed and implemented backend services and REST APIs that integrated with external systems, low-latency databases, and asynchronous processing pipelines.
- Introduced background task processing using Celery to support long-running and compute-heavy workloads, improving system responsiveness and reliability.
- Worked closely with frontend teams to deliver clean, performant user interfaces while ensuring seamless integration with backend and data services.
- Supported early machine learning and analytics features by building data ingestion pipelines and service abstractions used by downstream AI components.
- Collaborated with DevOps and platform teams on containerization, CI/CD pipelines, and cloud deployments, strengthening operational maturity across projects.

Software Developer

Centric Tech Inc. | October 2013 - June 2017

- Developed backend services and internal tools using Python and JavaScript to support business-critical systems and operational workflows.
 - Designed relational database schemas and optimized queries to support low-latency access patterns and growing data volumes.
 - Built and maintained APIs consumed by multiple internal applications, emphasizing correctness, maintainability, and clear service boundaries.
 - Participated in early distributed system designs, gaining hands-on experience with scalability challenges and production debugging.
 - Collaborated with senior engineers to improve code quality, introduce testing practices, and enhance system observability.
 - Supported production deployments and incident resolution, developing a strong foundation in reliability, monitoring, and long-term system ownership.
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